Project Report: V2V-Sync

Standalone Vehicle-to-Vehicle (V2V) Communication & Live Hazard Mapping

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1. Introduction

Vehicle-to-Vehicle (V2V) communication is a critical pillar of future Intelligent Transportation Systems (ITS). This project, **V2V-Sync**, focuses on creating a decentralized, infrastructure-less network that allows vehicles to exchange safety data. By bypassing traditional cellular networks, V2V-Sync ensures high-speed, low-latency alerts even in remote or disaster-hit areas.

2. Problem Statement

Current driver assistance systems rely heavily on onboard sensors (Cameras, LiDAR) which are limited by "line-of-sight." They cannot detect hazards around blind corners, in heavy fog, or beyond the vehicle immediately in front. Furthermore, manual reporting of road hazards like potholes is slow and inefficient.

3. Proposed System Architecture

The system consists of three main layers:

1. **Perception Layer:** Uses MPU6050 (Accelerometer) for pothole and accident detection and NEO-6M (GPS) for location tracking.
2. **Communication Layer:** Uses NRF24L01+ RF modules to broadcast Basic Safety Messages (BSM) to nearby vehicles using a mesh-capable protocol.
3. **Cloud Integration Layer:** A Python-based gateway pushes verified local hazards to a Firebase Realtime Database, which is then visualized on a live Google Maps dashboard.

4. Hardware Specifications

Component	Purpose
ESP32 / Arduino	Main processing unit for sensor fusion and logic.
NRF24L01+ PA/LNA	Long-range (up to 1km) wireless data transmission.
MPU6050	6-axis IMU for detecting sudden vertical jolts (potholes) or high-G impacts (accidents).
NEO-6M GPS	Provides sub-meter accuracy for coordinate tagging.

5. Software Logic & Incidents Covered

- **Pothole Detection:** The system monitors the Z-axis acceleration. A sudden deviation beyond a set threshold (e.g., $<4m/s^2$ or $>16m/s^2$) triggers a "Pothole Alert."
- **Accident Alert:** Sudden deceleration on the X or Y axis triggers an emergency broadcast to all trailing vehicles to prevent pile-ups.
- **Disaster/Block Mode:** Allows manual or sensor-based tagging of flooded roads or landslides, which is then daisy-chained through the network.

6. Live Google Maps Integration

Data from the V2V receiver is piped into a web-based dashboard. This dashboard uses the Google Maps JavaScript API to:

- Place dynamic markers based on incident type (Red for Accidents, Yellow for Potholes).
- Provide real-time updates without page refreshes.
- Allow transport authorities to view road health statistics remotely.

7. Conclusion & Future Scope

V2V-Sync demonstrates a cost-effective, scalable solution for road safety. Future iterations could include:

- **Li-Fi Integration:** Using vehicle headlights for optical V2V communication.
- **AI Integration:** Using machine learning to filter out "false positives" (like speed bumps) from actual potholes.